
Babylon.cpp Manual



Contents

1	Introduction	3
1.1	Supported Platforms	3
2	How It Works	3
2.1	Phonemization	3
2.2	TTS Engines	3
2.2.1	Kokoro	3
2.2.2	kittenTTS	4
2.2.3	VITS	4
3	Building from Source	4
3.1	Prerequisites	4
3.2	Cloning	4
3.3	Build Targets	4
3.4	Android	5
4	Models	5
4.1	Kokoro Voices	5
5	CLI	5
5.1	Global Options	5
5.2	Config File Format	7
5.3	phonemize	7
5.4	tts	7
5.5	serve	8
6	REST API	8
6.1	GET /status	8
6.2	GET /voices	8
6.3	GET /voices/kitten	9
6.4	POST /phonemize	9
6.5	POST /tts	9
7	C API	9
7.1	G2P (Phonemization)	10
7.1.1	babylon_g2p_init	10
7.1.2	babylon_g2p	10
7.1.3	babylon_g2p_tokens	10
7.1.4	babylon_g2p_free	10
7.2	VITS TTS	10
7.2.1	babylon_tts_init	10
7.2.2	babylon_tts	10
7.2.3	babylon_tts_free	11
7.3	Kokoro TTS	11
7.3.1	babylon_kokoro_init	11
7.3.2	babylon_kokoro_tts	11
7.3.3	babylon_kokoro_free	11
7.4	kittenTTS	11
7.4.1	babylon_kitten_init	11

7.4.2	babylon_kitten_tts	11
7.4.3	babylon_kitten_free	11
7.5	C API Example	12
8	C++ API	12
8.1	OpenPhonemizer::Session	12
8.2	Kokoro::Session	13
8.3	Kitten::Session	13
8.4	Vits::Session	13
8.5	C++ API Example	13
9	Python Wrapper	14
10	Web Frontend	14

1 Introduction

Babylon.cpp is a free and open-source C and C++ library for grapheme-to-phoneme (G2P) conversion and neural text-to-speech (TTS) synthesis. All inference runs locally using ONNX Runtime — no internet connection is required and no text or audio data leaves the host machine.

The library exposes three layers of interface:

- A **C API** suitable for use from any language with a C foreign-function interface.
- A **C++ API** providing higher-level session classes.
- A **command-line tool** (`babylon`) for phonemization, speech synthesis, and serving a REST API.

1.1 Supported Platforms

Platform	Architecture	Library
Linux	x86_64	libbabylon.so
macOS	Universal (x86_64 + arm64)	libbabylon.dylib
Windows	x86_64	babylon.dll
Android	arm64-v8a, x86_64	libbabylon.so

2 How It Works

2.1 Phonemization

The G2P pipeline converts raw text to IPA phonemes through three stages:

1. **Text normalisation** — Numbers are expanded to words, ordinals are resolved, abbreviations are handled, and punctuation is classified so the phonemizer receives clean prose.
2. **Dictionary lookup** — Each word is looked up in a bundled pronunciation dictionary (~130 000 English entries). If found, the dictionary phonemes are used directly, avoiding a model call.
3. **Neural G2P** — Words not found in the dictionary are phonemized by Open Phonemizer, an ONNX model that predicts IPA output character-by-character using CTC decoding.

2.2 TTS Engines

Babylon.cpp supports three neural TTS backends.

2.2.1 Kokoro

Kokoro is the recommended engine. It produces high-quality, multi-voice speech at 24 kHz mono. Each voice is represented by a 256-dimensional style embedding stored in a `.bin` file, indexed by the token count of the input sequence.

The Kokoro synthesis pipeline:

1. Phonemize the input text to an IPA string.
2. Encode the IPA string to Kokoro token IDs using a built-in 178-entry vocab.
3. Load the voice style embedding for the current token count from the `.bin` file.

4. Run the Kokoro ONNX model with `input_ids`, `style`, and `speed` inputs to produce a PCM waveform.
5. Write the waveform as a 24 kHz WAV file.

2.2.2 kittenTTS

kittenTTS reuses the same Open Phonemizer pipeline and Kokoro-compatible IPA vocabulary, but applies a slightly different input wrapping and voice-style lookup strategy. Each voice is stored as a `.bin` file of float32 style rows with dimension 256.

The kittenTTS synthesis pipeline:

1. Phonemize the input text to an IPA string.
2. Encode the IPA string to token IDs using the Kokoro-compatible vocab, then append the kittenTTS stop marker so the final input becomes `[0, ..., 10, 0]`.
3. Load the voice style embedding row indexed by the phoneme-string length.
4. Run the kittenTTS ONNX model with `input_ids`, `style`, and `speed` inputs.
5. Trim the short trailing artefact region and write the waveform as a 24 kHz WAV file.

2.2.3 VITS

VITS is an end-to-end neural TTS model. Piper-compatible VITS models are supported. The sample rate is determined by metadata embedded in the model file.

3 Building from Source

3.1 Prerequisites

- CMake 3.18 or later
- A C++17 compiler (GCC, Clang, or MSVC)
- Git (for submodule checkout)
- **macOS only:** Xcode Command Line Tools
- **Windows only:** Visual Studio 2019 or later with the C++ workload

3.2 Cloning

```
git clone --recursive \
  https://github.com/Mobile-Artificial-Intelligence/babylon.cpp.git
cd babylon.cpp
```

3.3 Build Targets

Target	Description
<code>make lib</code>	Build <code>libbabylon</code> only
<code>make cli</code>	Build the library, CLI binary, and install runtime files into <code>bin/</code>
<code>make install</code>	Install the same runtime layout into <code>INSTALL_DIR</code>
<code>make debug</code>	CLI build in Debug mode
<code>make android</code>	Cross-compile the library for Android (requires NDK)

All build output is placed in `bin/`. The source-root `bin/` tree and `cmake -install` share the same layout: `bin/babylon` for the executable, `bin/lib/` for shared libraries, `bin/data/` for config and web assets, `bin/models/` for ONNX models, and `bin/voices/` for Kokoro and kittenTTS voice files.

3.4 Android

Android builds require the Android NDK (r27 or later). Set `ANDROID_NDK_HOME` before running `make android`.

4 Models

The library requires external model files and voice assets that are not bundled in the repository. Place model files in `models/`, voice files in `voices/`, or configure their paths in `data/config.json`.

File	Description	Config Key
<code>open-phonemizer.onnx</code>	Open Phonemizer G2P model	<code>phonemizer_model</code>
<code>dictionary.json</code>	Pronunciation dictionary	<code>dictionary</code>
<code>kokoro-quantized.onnx</code>	Kokoro TTS model	<code>kokoro_model</code>
<code>voices/kokoro/*.bin</code>	Kokoro voice style files	<code>kokoro_voices</code>
<code>kitten-tts.onnx</code>	kittenTTS model	<code>kitten_model</code>
<code>voices/kitten/*.bin</code>	kittenTTS voice style files	<code>kitten_voices</code>
<code>en-US-curie-vits.onnx</code>	VITS TTS model	<code>vits_model</code>

4.1 Kokoro Voices

Each voice is a `.bin` file of float32 values. The filename without extension is used as the voice name. The default voice is `en-US-heart`.

Voice names follow the convention `<lang>-<name>`:

Language	Example Voice Names
English (US)	<code>en-US-heart</code> , <code>en-US-bella</code> , <code>en-US-nova</code> , <code>en-US-adam</code> , ...
English (GB)	<code>en-GB-alice</code> , <code>en-GB-emma</code> , <code>en-GB-daniel</code> , ...
German	<code>de-DE-dora</code> , <code>de-DE-alex</code>
French	<code>fr-FR-siwis</code>
Japanese	<code>ja-JP-alpha-f</code> , <code>ja-JP-kumo</code> , ...
Chinese (Simplified)	<code>zh-CN-xiaobei</code> , <code>zh-CN-yunxi</code> , ...

5 CLI

The `babylon` binary provides three subcommands: `phonemize`, `tts`, and `serve`.

5.1 Global Options

The following options apply to all subcommands and are processed before dispatch:

Option	Argument	Description
-config	<path>	Load a JSON config file
-phonemizer-model	<path>	Phonemizer ONNX model
-dictionary	<path>	Pronunciation dictionary JSON
-kokoro-model	<path>	Kokoro ONNX model
-kokoro-voice	<name>	Default Kokoro voice name
-kokoro-voices	<dir>	Directory of voice .bin files
-kitten-model	<path>	kittenTTS ONNX model
-kitten-voice	<name>	Default kittenTTS voice name
-kitten-voices	<dir>	Directory of kittenTTS .bin files
-vits-model	<path>	VITS ONNX model
-h, -help		Show help

On startup, `babylon` automatically looks for `data/config.json` under the installed tree and loads it silently. A `-config` flag overrides this, and individual flags override specific keys.

5.2 Config File Format

```
{
  "phonemizer_model": "../models/open-phonemizer.onnx",
  "dictionary": "dictionary.json",
  "kokoro_model": "../models/kokoro-quantized.onnx",
  "kokoro_voice": "en-US-heart",
  "kokoro_voices": "../voices/kokoro",
  "kitten_model": "../models/kitten-tts.onnx",
  "kitten_voice": "en-US-bella",
  "kitten_voices": "../voices/kitten",
  "vits_model": "../models/en-US-curie-vits.onnx",
  "host": "127.0.0.1",
  "port": 8775
}
```

5.3 phonemize

Convert text to IPA phonemes.

```
babylon phonemize "Hello_world"
babylon phonemize --tokens "Hello_world"
```

Option	Description
-tokens	Print Kokoro token IDs instead of the IPA string
-h, -help	Show help

5.4 tts

Synthesise speech and write a WAV file.

```
babylon tts "Hello_world"
babylon tts "Hello_world" -o hello.wav
babylon tts --voice en-US-nova --speed 1.2 "Hello_world"
babylon tts --kitten --voice en-US-bella "Hello_world"
babylon tts --vits "Hello_world" -o hello.wav
```

Option	Argument	Description
-kokoro		Use the Kokoro engine (default)
-kitten		Use the kittenTTS engine
-vits		Use the VITS engine
-engine	<name>	Select kokoro, kitten, or vits explicitly
-v, -voice	<name>	Voice name for Kokoro or kittenTTS
-kokoro-voice	<name>	Same as -voice for Kokoro
-kitten-voice	<name>	Same as -voice for kittenTTS
-speed	<float>	Speech speed multiplier for Kokoro or kittenTTS (default: 1.0)
-o	<path>	Output WAV file (default: output.wav)
-timings	<path>	Write timing JSON alongside the WAV
-h, -help		Show help

Voice names are filenames in the engine-specific voices directory without the `.bin` extension. For example, `-voice en-US-heart` maps to `voices/kokoro/en-US-heart.bin`, while `-kitten -voice en-US-bella` maps to `voices/kitten/en-US-bella.bin`.

5.5 serve

Start a local REST API server with the web frontend.

```
babylon serve
babylon serve --host 0.0.0.0 --port 9000
```

Option	Argument	Description
-host	<addr>	Bind address (default: 127.0.0.1)
-port	<port>	Port number (default: 8775)
-h, -help		Show help

On startup, all configured models are pre-loaded. The web frontend is served at `GET /` from `data/index.html` under the installed tree.

6 REST API

When running `babylon serve`, the following HTTP endpoints are available. All responses include `Access-Control-Allow-Origin: *`.

6.1 GET /status

Returns the availability of each engine and the number of loaded voices.

Response (application/json):

```
{
  "phonemizer": true,
  "kokoro":     true,
  "kitten":     true,
  "vits":       false,
  "voices":     54,
  "kitten_voices": 8
}
```

`phonemizer`, `kokoro`, `kitten`, and `vits` are booleans indicating whether the model file exists at the configured path. `voices` is the Kokoro voice count, and `kitten_voices` is the kittenTTS voice count.

6.2 GET /voices

Returns a sorted JSON array of available Kokoro voice names.

Response (application/json):

```
["en-GB-alice", "en-US-bella", "en-US-heart", ...]
```

6.3 GET /voices/kitten

Returns a sorted JSON array of available kittenTTS voice names.

Response (application/json):

```
["en-US-bella", "en-US-bruno", "en-US-hugo", ...]
```

6.4 POST /phonemize

Convert text to IPA phonemes or Kokoro token IDs.

Request body (application/json):

Field	Type	Required	Description
text	string	Yes	Input text to phonemize
tokens	boolean	No	Return token IDs instead of IPA (default: <code>false</code>)

Response — IPA mode:

```
{ "phonemes": "<IPA string>" }
```

Response — token mode:

```
{ "tokens": [31, 29, 42, 0, 51, 17, 32, 42] }
```

6.5 POST /tts

Synthesise speech. Returns a WAV audio file on success.

Request body (application/json):

Field	Type	Required	Description
text	string	Yes	Input text
engine	string	No	"kokoro" (default), "kitten", or "vits"
voice	string	No	Voice name; defaults to the config value
speed	number	No	Speech speed multiplier for Kokoro or kittenTTS (default: 1.0)

Success response: audio/wav binary body.

Error response (application/json):

```
{ "error": "description of the error" }
```

7 C API

Include `babylon.h` and link against `libbabylon`.

7.1 G2P (Phonemization)

7.1.1 babylon_g2p_init

```
int babylon_g2p_init(const char* model_path,
                    babylon_g2p_options_t options);
```

Loads the Open Phonemizer ONNX model. Returns 0 on success, non-zero on failure.

Options struct:

```
typedef struct {
    const char*      dictionary_path; // path to dictionary.json, or
    NULL
    const unsigned char use_punctuation; // 1 to preserve punctuation, 0
    to strip
} babylon_g2p_options_t;
```

7.1.2 babylon_g2p

```
char* babylon_g2p(const char* text);
```

Phonemizes `text` and returns a heap-allocated IPA string. The caller must call `free()` on the result.

7.1.3 babylon_g2p_tokens

```
int* babylon_g2p_tokens(const char* text);
```

Phonemizes `text` and returns a heap-allocated, -1-terminated array of Kokoro-compatible token IDs. The caller must call `free()` on the result.

7.1.4 babylon_g2p_free

```
void babylon_g2p_free(void);
```

Releases the G2P session and frees all associated memory.

7.2 VITS TTS

7.2.1 babylon_tts_init

```
int babylon_tts_init(const char* model_path);
```

Loads a VITS ONNX model. The G2P session must already be initialised. Returns 0 on success.

7.2.2 babylon_tts

```
void babylon_tts(const char* text, const char* output_path);
```

Synthesises `text` and writes a WAV file to `output_path`.

7.2.3 babylon_tts_free

```
void babylon_tts_free(void);
```

Releases the VITS session.

7.3 Kokoro TTS

7.3.1 babylon_kokoro_init

```
int babylon_kokoro_init(const char* model_path);
```

Loads the Kokoro ONNX model. The G2P session must already be initialised. Returns 0 on success.

7.3.2 babylon_kokoro_tts

```
void babylon_kokoro_tts(const char* text,
                        const char* voice_path,
                        float speed,
                        const char* output_path);
```

Synthesises `text` using the voice style loaded from `voice_path` at the given `speed`, writing a WAV file to `output_path`.

7.3.3 babylon_kokoro_free

```
void babylon_kokoro_free(void);
```

Releases the Kokoro session.

7.4 kittenTTS

7.4.1 babylon_kitten_init

```
int babylon_kitten_init(const char* model_path);
```

Loads the kittenTTS ONNX model. The G2P session must already be initialised. Returns 0 on success.

7.4.2 babylon_kitten_tts

```
void babylon_kitten_tts(const char* text,
                        const char* voice_path,
                        float speed,
                        const char* output_path);
```

Synthesises `text` using the kittenTTS voice style loaded from `voice_path` at the given `speed`, writing a WAV file to `output_path`.

7.4.3 babylon_kitten_free

```
void babylon_kitten_free(void);
```

Releases the kittenTTS session.

7.5 C API Example

```
#include "babylon.h"
#include <stdlib.h>

int main(void) {
    babylon_g2p_options_t opts = {
        .dictionary_path = "data/dictionary.json",
        .use_punctuation = 1,
    };

    if (babylon_g2p_init("models/open-phonemizer.onnx", opts) != 0)
        return 1;

    if (babylon_kokoro_init("models/kokoro-quantized.onnx") != 0)
        return 1;

    babylon_kokoro_tts(
        "Hello_world",
        "voices/kokoro/en-US-heart.bin",
        1.0f,
        "output.wav"
    );

    babylon_kokoro_free();
    babylon_g2p_free();
    return 0;
}
```

8 C++ API

Include `babylon.h` and link against `libbabylon`. All classes are in their respective namespaces.

8.1 OpenPhonemizer::Session

```
namespace OpenPhonemizer {
    class Session {
    public:
        Session(const std::string& model_path,
                const std::string& dictionary_path = "",
                bool use_punctuation = false);

        // Returns concatenated IPA phoneme string for full text
        std::string phonemize(const std::string& text);

        // Returns Kokoro-compatible token IDs
        std::vector<int64_t> phonemize_tokens(const std::string& text);
    };
}
```

8.2 Kokoro::Session

```
namespace Kokoro {
    class Session {
    public:
        Session(const std::string& model_path);

        void tts(const std::string& phonemes,
                 const std::string& voice_path,
                 float speed,
                 const std::string& output_path);
    };
}
```

8.3 Kitten::Session

```
namespace Kitten {
    class Session {
    public:
        Session(const std::string& model_path);

        void tts(const std::string& phonemes,
                 const std::string& voice_path,
                 float speed,
                 const std::string& output_path);
    };
}
```

8.4 Vits::Session

```
namespace Vits {
    class Session {
    public:
        Session(const std::string& model_path);

        void tts(const std::vector<std::string>& phonemes,
                 const std::string& output_path);
    };
}
```

8.5 C++ API Example

```
#include "babylon.h"

int main() {
    OpenPhonemizer::Session phonemizer(
        "models/open-phonemizer.onnx",
        "data/dictionary.json",
        /* use_punctuation = */ true
    );
}
```

```

Kokoro::Session kokoro("models/kokoro-quantized.onnx");

std::string phonemes = phonemizer.phonemize("Hello_world");

kokoro.tts(phonemes,
            "voices/kokoro/en-US-heart.bin",
            /* speed = */ 1.0f,
            "output.wav");

return 0;
}

```

9 Python Wrapper

A pre-built Python package is available as a CI artifact. It bundles the compiled libraries for Linux, macOS, and Windows alongside the Python wrapper module (`babylon.py`).

```

import babylon

babylon.g2p_init("models/open-phonemizer.onnx",
                "data/dictionary.json")
babylon.kokoro_init("models/kokoro-quantized.onnx")

babylon.kokoro_tts(
    "Hello_world",
    "voices/kokoro/en-US-heart.bin",
    speed=1.0,
    output_path="output.wav"
)

babylon.kokoro_free()
babylon.g2p_free()

```

10 Web Frontend

When running `babylon serve`, a single-page web interface is served at `http://<host>:<port>/`. It requires no additional dependencies and communicates entirely with the local REST API.

Features:

- **Status indicator** — A dot in the header reflects engine availability from `GET /status`. Cyan indicates at least one engine is ready; red indicates no models are configured.
- **Engine selector** — Switches between Kokoro, kittenTTS, and VITS. Options are disabled if the corresponding model is not available.
- **Voice selector** — Populated from `GET /voices` or `GET /voices/kitten`; disabled when VITS is selected.
- **Speed slider** — Controls Kokoro and kittenTTS speech speed from $0.5\times$ to $2.0\times$.
- **Phonemize** — Sends `POST /phonemize` and displays the IPA result.
- **Speak** — Sends `POST /tts` and plays the returned WAV audio directly in the browser.
- **Keyboard shortcut** — `Ctrl+Enter` / `Cmd+Enter` triggers speech synthesis.